

Supervision Meeting Brief

Progress report and decisions required

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1. Summary

The data pipeline is complete. A ten year daily PM2.5 series for Delhi has been acquired, audited, cleaned and converted into a supervised feature table, and the code that does so is tested and reproducible. This was the whole of the first planned week and it is finished ahead of schedule.

Two problems were found in the source data during the audit, one of which was silently corrupting the series, and both are documented in Section 3. One assumption in the approved proposal has been contradicted by the data and the model specification changes as a result, which is set out in Section 4. Four decisions require your input and are listed in Section 5.

2. Work completed

2.1 Station selection

Five candidate monitoring stations within 25 km of central Delhi were compared on data coverage rather than on nominal length of record. Coverage is the share of the calendar carrying a usable daily mean, and gap is the longest unbroken run of days with no usable value.

Location	Provider	First	Years	Coverage	Gap	Exceed
17, R K Puram	CPCB	2025-02-19	1.5	85.9%	11	21.2%
50, Punjabi Bagh	CPCB	2025-02-19	1.5	85.2%	11	20.7%
235, Anand Vihar	CPCB	2025-02-19	1.5	86.8%	10	23.9%
8118, New Delhi	AirNow	2016-11-09	9.8	89.4%	31	30.0%
5404, Pusa	CPCB	2025-02-19	1.5	76.2%	10	15.1%

The decisive result is in the third column. Every CPCB station returns measurements only from 19 February 2025, roughly eighteen months, even though the OpenAQ location metadata advertises a first observation of 5 February 2016. Four independent stations sharing an identical cutoff is a systematic limit on what the measurements endpoint serves rather than a coincidence. Only station 8118 returns the full record, so it was selected.

This carries a cost that must be stated in the report. Station 8118 is supplied through AirNow and is almost certainly the United States diplomatic post monitor rather than part of the Indian official network. The project therefore forecasts at one well characterised reference site in Delhi, which is common in this literature, and cites OpenAQ and the United States Department of State rather than CPCB.

2.2 Two defects in the source data

Physically impossible values. The raw series contained five negative concentrations, to a minimum of -7.3, and a maximum of 1990. The value 1990.0 appeared three times, identically, on unrelated dates, which is the signature of a clipped instrument ceiling rather than a measurement. One of the three fell on 2 September, in the monsoon, between neighbouring days of 42 and 178.

A wrong coverage denominator. OpenAQ reports the completeness of each daily aggregate as the observation count divided by an expected count of 24, assuming hourly reporting. This sensor reported half hourly from 2016 to 2024 and hourly from 2025. The visible symptom is a reported completeness above 100 percent, which is impossible. The damaging symptom is invisible: a day holding 24 of a real 48 readings is scored as fully complete, so a mean computed from half a day of a polluted afternoon enters the series as though it were a genuine daily mean. Days at 50 to 75 percent true coverage were approximately nine times more likely to exceed 500 than days above 90 percent, which identifies this as the mechanism producing the impossible extremes.

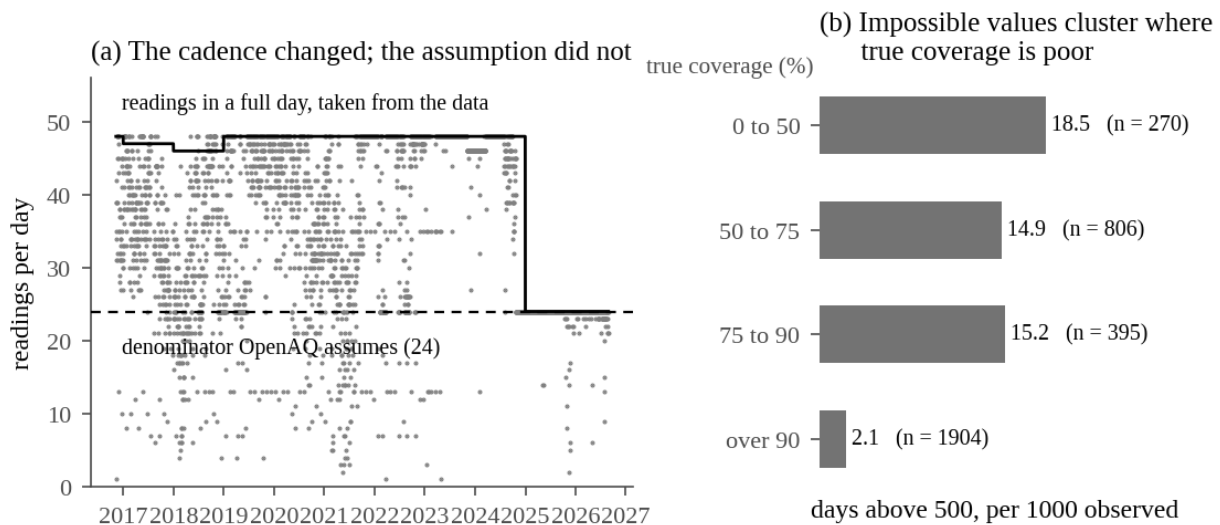


Figure 1. The coverage defect. Panel (a) shows the number of readings contributing to each daily mean, with the cadence estimated from the data each year as the solid line. The sensor reported roughly half hourly until the end of 2024 and hourly thereafter, while the denominator used to report completeness stayed at 24 throughout. Panel (b) shows the rate of physically implausible values by true coverage. Days observed for more than 90 percent of their length yield such values at roughly one seventh the rate of days below that, which identifies incomplete days as the mechanism rather than a coincidence.

The correction takes the reporting cadence from the data itself, using the ninetieth percentile of observation counts within each year, which follows the 2025 change in reporting interval automatically. Recomputed on that basis, 1,085 of 3,199 observed days fall below 75 percent coverage rather than the 268 the published figure implies.

2.3 Cleaning applied

Rule	Days removed	Basis
Value below zero	5	Mass concentration cannot be negative
Coverage below 50 percent	270	Against the true cadence, not the published one
Value above 1000	7	Judgement, see Section 5
Total	282 of 3,375	

The coverage threshold is set at 50 rather than a stricter value because the longest gap is the binding constraint. At 60 percent the worst gap rises to 69 days and at 90 percent to 185 days, either of which would distort the estimation of the annual cycle. At 50 percent the worst gap is 34 days. The

exceedance rate is stable across every candidate threshold, moving only from 30.0 to 27.0 percent, which confirms the filter is not selectively removing the events of interest.

2.4 The resulting dataset

Property	Value
Source	OpenAQ location 8118, sensor 23534, provider AirNow
Period	9 November 2016 to 27 August 2026, 9.8 years
Usable daily means	3,093, being 86.4 percent of the calendar
Longest gap	34 consecutive days
Mean, median, maximum	102.7, 71.7, 896.0
Days above 121	920, being 29.7 percent

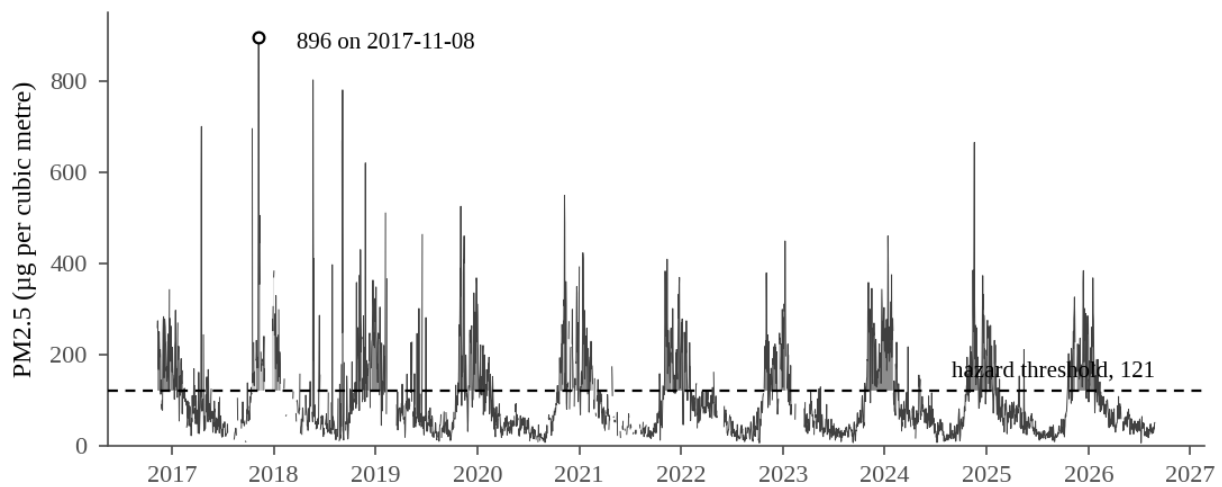


Figure 2. The cleaned daily series. Shading marks the days exceeding the threshold. The episodic structure is the feature the project exists to forecast: exceedances do not arrive as isolated days but as multi day winter episodes, which is why persistence of the event itself is carried as a predictor in the feature table.

The maximum of 896 falls on 8 November 2017, within the documented severe episode of that month. A real event surviving the cleaning while the September and February spikes do not is a useful check that the rules are discriminating rather than merely truncating.

2.5 Feature table

3,355 rows, 39 feature columns and 6 target columns. A row indexed by date t is one forecast origin, and every feature in it is a function of observations up to and including day t . Features comprise lags at 0, 1, 2, 3 and 7 days, first differences, rolling mean, standard deviation and maximum over 3, 7, 14 and 30 days, an anomaly and standardised anomaly against the 30 day norm, exceedance frequency over the last 7 and 30 days, days since the last exceedance, calendar terms, four pairs of annual Fourier terms and three missingness indicators.

Targets are the concentration at one, two and three days ahead and the matching binary exceedance indicators, so both arms of Research Question 2 are answered from a single table. Class balance is 29.7 percent at all three horizons.

An automated leakage test accompanies the construction. It rebuilds the table from a series corrupted from a chosen cut date onward and asserts that every feature row before that cut is unchanged, then verifies target alignment and the right edge alignment of the rolling windows. It runs before every build and currently passes.

3. A finding that changes the model specification

The approved proposal states that daily pollution series carry a weekly cycle driven by traffic and industrial activity, and the accompanying plan treats the coexistence of a weekly and an annual period as the central modelling difficulty, since a seasonal ARIMA accommodates only one seasonal period. That difficulty was to be the main question for this meeting.

Tested on the cleaned series, the weekly cycle is not present at this station. Three independent tests agree.

Test	Result
Periodogram, dominant periods	357.9, 178.9, 188.4, 397.7, 325.4, 60.7 days. The annual cycle and its harmonics. Nothing near 7.
Autocorrelation of the deseasonalised log series	Lag 6 is 0.179, lag 7 is 0.173, lag 8 is 0.153. Lag 7 lies on the decay curve rather than above it. The same holds at lags 14 and 21.
Day of week effect on the same residuals	Spread of 6.5 percent between the highest and lowest day. ANOVA $F = 1.13$, $p = 0.34$. Kruskal-Wallis $p = 0.77$.

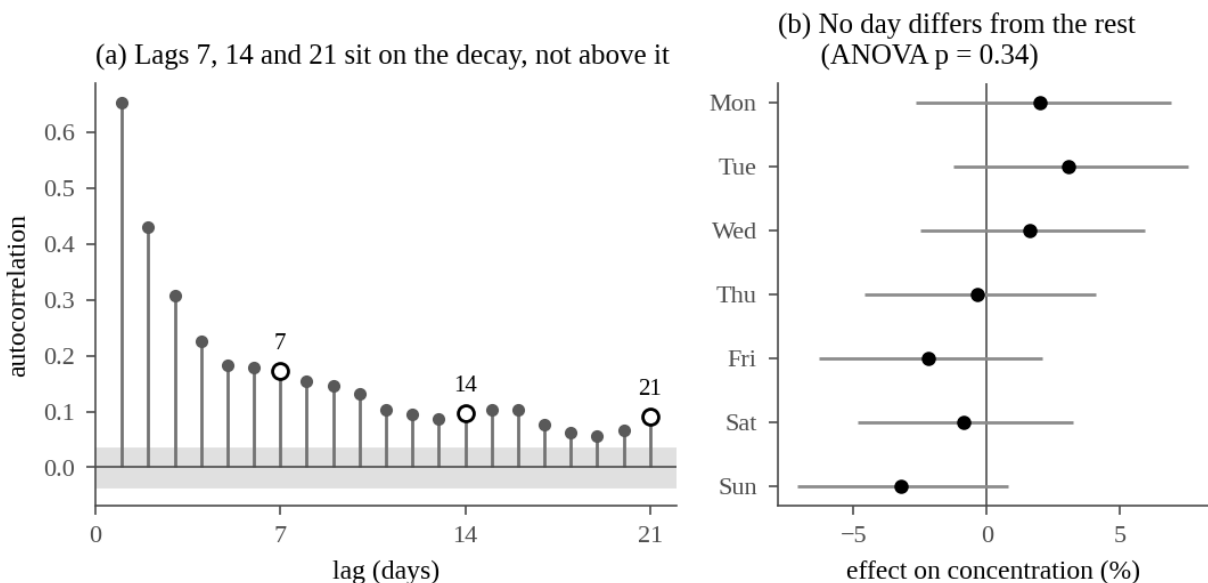


Figure 3. No weekly cycle. Panel (a) is the autocorrelation of the deseasonalised logarithm of concentration; lags 7, 14 and 21 are circled and lie on the decay curve rather than above it, which is persistence rather than periodicity. The shaded band is the 95 percent interval for zero. Panel (b) gives the effect of each day of the week on the same residuals with 95 percent intervals; every interval contains zero.

There is a coherent physical reading rather than a null result. A weekly rhythm in urban particulate concentration is a traffic signature, and station 8118 sits in the diplomatic enclave rather than at a roadside site such as ITO or Anand Vihar. Its absence at a site that is not traffic dominated is what one would expect.

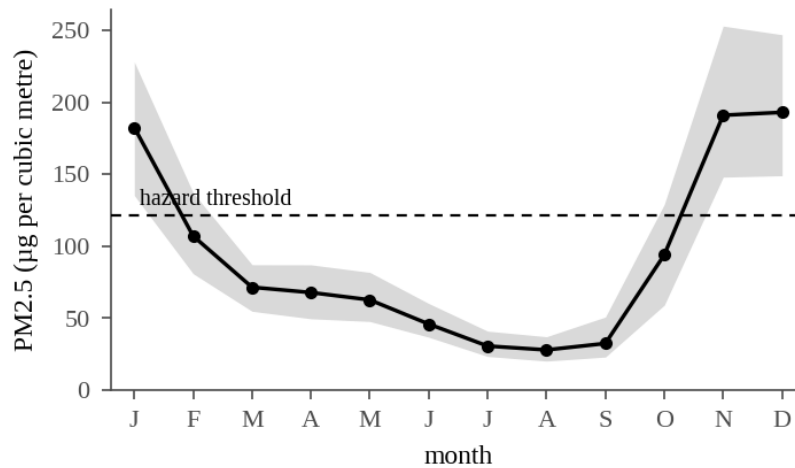


Figure 4. The annual cycle, by contrast, is unmistakable. Monthly median with the interquartile range shaded. The median exceeds the hazard threshold in November, December and January and falls to roughly a sixth of that in the monsoon. This is the structure the Fourier terms are there to capture, and its strength is the reason the absence in Figure 3 is credible rather than an artefact of a weak test.

Consequence. The classical benchmark becomes ARIMA with annual Fourier terms as exogenous regressors, with no weekly seasonal order. The specification is simpler, better identified, and now justified empirically rather than assumed. Two further points follow. Annual Fourier terms explain 69 percent of the variance of the logarithm of concentration against considerably less on the raw scale, so the classical modelling will be done in logs. Four Fourier pairs are used; the Bayesian information criterion continues to improve to six, but it assumes independent errors and these residuals are strongly autocorrelated, so it will over-select, and the order will be treated as a tuning parameter under the rolling origin evaluation instead.

4. Evaluation design

The models are the straightforward part of what remains. The evaluation is where the project can quietly fail, so it is worth setting out now rather than at the end.

Rolling origin rather than a single split. A model will be fitted on an expanding window, used to forecast one, two and three days ahead, then refitted with the origin moved forward. Every forecast is therefore made using only information that existed at the moment it was issued. This is the fiddliest code in the project and the easiest place to leak future information into past forecasts, so it will be built and tested on a short window before being run at length, and it will carry the same corruption test already used on the feature table: a series is corrupted from a chosen date onward, and any forecast issued before that date must be unchanged.

Metrics reported separately at each horizon. Mean absolute error, root mean squared error, and mean absolute scaled error against the naive forecast. Skill decays with horizon and averaging across horizons would conceal exactly that.

The warning system, which is the point. A model with good average error can still miss the extreme days, and those are the only days anyone acts on. For each approach the decision threshold will be swept from one extreme to the other, and at each point the hit rate, the false alarm rate, the precision, and the number of alarms raised per year will be recorded. That last quantity is what an operator actually lives with and it is rarely reported. The resulting curve of hit rate against false alarm rate, with the benchmark operating points marked on it, is the figure the report is built around.

Class imbalance handled explicitly. Exceedances are a minority at 29.7 percent and would be a much smaller minority under the severe threshold. Class weighting and precision-recall analysis will be used rather than accuracy, and the minority class will not be resampled, since duplicating days

breaks the temporal structure the whole design depends on.

5. Decisions required

- **1. The hazardous threshold.** The scripts currently use 121 microgrammes per cubic metre, the CPCB breakpoint at which the category becomes very poor, which yields an event rate of 29.7 percent. The severe breakpoint at 250 would yield a far smaller and more extreme event set with correspondingly wider confidence intervals on every hit rate. The choice determines what the warning system is for and it should be settled before modelling begins.
- **2. Acceptability of the station provenance.** Using a United States diplomatic post monitor rather than the Indian official network is forced by the CPCB truncation described in Section 2.1. Whether this is acceptable, and how prominently it should be caveated, is a judgement I would rather take from you than make alone. A secondary route through the OpenAQ bulk archive may hold the full CPCB record and is worth an hour of investigation in week 6 as a robustness check.
- **3. The upper cleaning threshold.** Removing values above 1000 is the weakest assumption in the pipeline. It rests on seasonal implausibility rather than on a documented instrument limit. The intention is to rerun the complete analysis without that rule in week 5 and report whether any conclusion changes, which converts the assumption into a stated robustness check. Please confirm that is sufficient.
- **4. Scope of the classical comparison.** Given that the two seasonal period problem does not arise, the structural state space model is no longer required to resolve it. Would you still like it fitted as a decomposition comparison, or is the effort better spent on the machine learning suite and the evaluation?

6. Work remaining

Submission is due 22 October 2026. The schedule below front loads the modelling so that all results are frozen by 30 September, leaving two weeks for writing and one week of contingency.

Wee k	Dates	Work	Completion criterion
1	20 to 26 Aug	Data acquisition, cleaning, feature table	Complete, 20 August
2	27 Aug to 2 Sep	Naive, seasonal naive and climatology benchmarks; ARIMA with Fourier exogenous terms on log concentration	Error metrics at each horizon for every benchmark
3	3 to 9 Sep	Rolling origin evaluation harness	Benchmarks reproduced under the harness with a passing leakage test
4	10 to 16 Sep	Penalised regression, random forest, histogram gradient boosting	Research Question 1 answered
5	17 to 23 Sep	Direct exceedance classification and threshold sweep	Research Questions 2 and 3 answered; hit rate against false alarm rate produced
6	24 to 30 Sep	Recurrent network; robustness checks; results frozen	No further modelling after 30 September
7	1 to 7 Oct	First complete draft	Draft to supervisor
8	8 to 14 Oct	Revision on supervisor feedback	Second draft, figures final

7. Reproducibility

Four command line scripts carry the pipeline from an empty folder to the feature table: acquisition from the OpenAQ interface, a viability and structure check, the cleaning step, and feature construction. Each accepts a synthetic demonstration mode and each carries an automated correctness test. The cleaning script alters nothing unless explicitly instructed, and reports what each rule would remove before it is applied, so every threshold in Section 2.3 is a recorded decision rather than a default. Raw and cleaned data are retained under version control so that the analysis does not depend on the data provider serving identical values at a later date, which the CPCB finding suggests is not a safe assumption.